Olist Growth and Delivery Priorities

Executive summary January 2017 to August 2018

Business question Where should Olist focus operational attention as delivered order volume expands?

Recommendation Prioritise delivery diagnosis in SP and RJ, while running a separate investigation of low ratings on orders that arrived on time.

Delivered item sales	Delivered orders	Late-delivery rate
BRL 13.18M	96,211	6.8%

Three findings

Growth is driven by orders. January-August sales increased 141.1% between 2017 and 2018; orders increased 139.9%, while AOV rose only 0.5%. More sales therefore require substantially more fulfilment activity.

Delivery problems are concentrated. SP and RJ account for 50.7% of late orders. SP has the largest number of late orders; RJ combines substantial volume with an elevated late rate.

Late delivery is not the whole ratings problem. Late orders have a 62.4% low-rating rate, but 67.5% of all low ratings in the valid-delivery cohort come from on-time or early orders.

Two priorities

Priority	Immediate next step
Delivery diagnosis	Compare sellers, routes and dispatch versus transit stages in SP and RJ, including abnormal purchase cohorts.
On-time dissatisfaction	Classify low-rating review text by issue, category and seller before selecting a customer-experience intervention.

Analysis scope Historical purchase dates from 1 January 2017 to 31 August 2018. Sales exclude freight and cover delivered orders; delivery metrics use valid records. These are diagnostic priorities, not measured intervention effects.

Growth is driven by order volume

Compare the same eight months before interpreting growth

Metric	2017 Jan-Aug	2018 Jan-Aug	Change
Delivered item sales	BRL 2.99M	BRL 7.22M	+141.1%
Delivered orders	21,998	52,783	+139.9%
Item AOV	BRL 136.08	BRL 136.75	+0.5%

Sales and order counts increased at almost the same rate. With **sales = orders x AOV**, the aggregate increase is predominantly volume-driven. This does not establish whether additional orders came from new customers or repeat purchases.

Business implication Dispatch, transport and support capacity should be planned around order throughput. Sales growth without much AOV growth adds transactions to fulfil rather than materially more revenue per transaction.

Concentration helps locate the workload

Full-period segment	Share of delivered item sales
Sao Paulo state	38.4%
SP, RJ and MG combined	63.4%
Top 10 product categories	62.5%

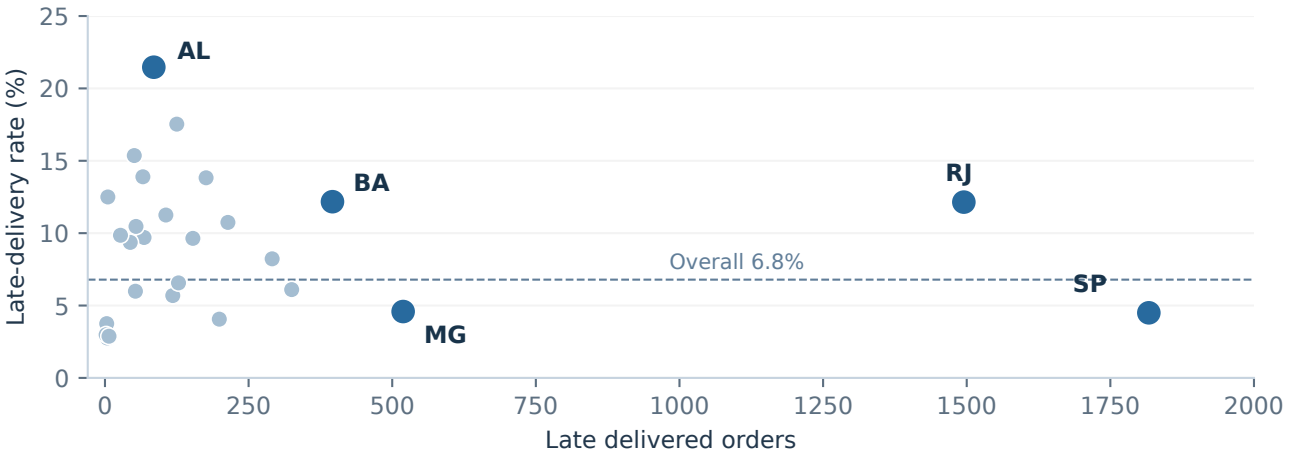
Use these segments to monitor demand and service capacity. Concentration alone does not identify profitability or justify higher marketing spend.

Next step

Distinguish new and returning customers using customer_unique_id, then examine item quantities, prices and category mix to explain AOV. Keep the same-period comparison when assessing growth.

Prioritise delivery by rate and scale

SP and RJ need different diagnostic questions



Dashed line: overall late rate. Each point is a state; sample sizes differ. SP = Sao Paulo; RJ = Rio de Janeiro; AL = Alagoas; BA = Bahia; MG = Minas Gerais.

SP: 1,817 late orders at a 4.5% late rate. **RJ:** 1,495 at 12.1%. Together they represent 50.7% of all late orders. AL has the highest rate, 21.5%, but only 85 late orders among 396 valid deliveries.

Business implication SP offers a large pool of affected orders to investigate; RJ merits diagnosis of its elevated failure rate. Improvement budgets should not be prioritised by late rate alone.

Purchase cohort	Valid deliveries	Late orders	Late rate
Feb-Mar 2018	13,558	2,254	16.6%
Apr-May 2018	13,547	749	5.5%

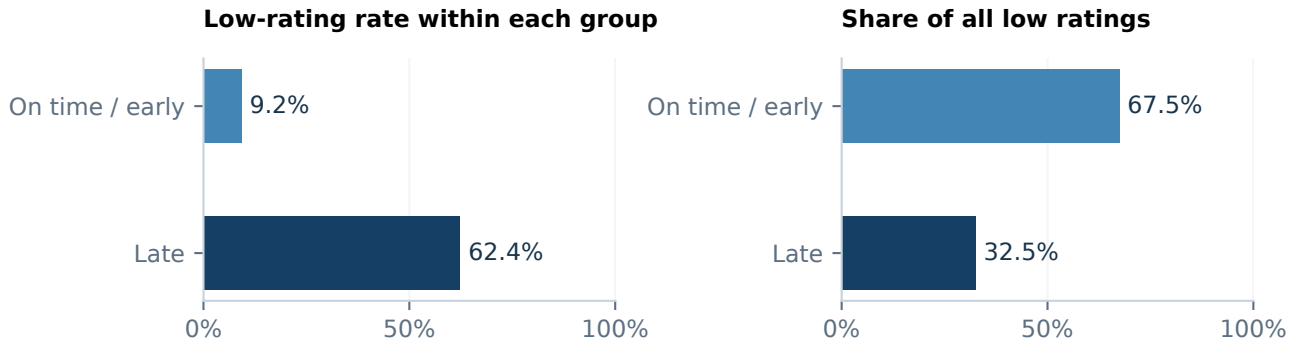
Nearly identical order volumes have very different late rates. Volume alone is insufficient to explain the gap; seller mix, geography, routes and promised windows need comparison.

Next step Decompose dispatch and transit stages within comparable segments, starting with SP, RJ and the higher-delay purchase cohorts.

Purchase cohorts describe when orders were placed, not when a disruption occurred. Geographic and time comparisons are unadjusted.

Low ratings extend beyond late delivery

Separate the risk within a group from its share of all low ratings



Left: low ratings / rated orders in each group (on time n=89,182; late n=6,378). Right: group low ratings / all 12,228 low ratings with valid deliveries. Low rating = 1 or 2 stars.

Late orders are much more likely to receive low ratings. However, on-time or early orders contribute 8,248 low ratings, or 67.5% of the total, because that group is much larger.

Business implication Late delivery and on-time dissatisfaction warrant parallel investigation. The latter requires review-text and product/seller analysis before attributing causes.

Review timing changes the comparison

Delay group	All valid reviews	Submitted after delivery
Late 8-14 days	1.67 / 5 n = 1,446	3.53 / 5 n = 57
Late 15+ days	1.72 / 5 n = 1,333	2.89 / 5 n = 19

Excluding reviews submitted while customers waited removes many severe-delay observations. The remaining small subsets answer a different question; they are not inherently more representative.

Next step Classify on-time low-rating reviews and validate issue labels manually. Enhanced communication for unresolved delays should be evaluated through a pilot.

The timing comparison uses different subsets, not before-and-after ratings from the same customers. Small sample sizes must remain visible.

Actions and analytical boundaries

Translate the findings into diagnosis and measurable tests

Action	Validation measure
Diagnose SP and RJ by seller, route and fulfilment stage. Assess cost and controllability before allocating resources.	Late rate, affected count and severe-delay share within comparable segments.
Classify on-time low-rating reviews; review labels manually and compare categories and sellers.	Issue frequencies, affected orders and a stated rated-order denominator.
Pilot follow-up for eligible unresolved delays using an appropriate comparison group.	Predefined ratings, complaints and service costs; assess allocation bias.

Methods and definitions

Data preparation. The workflow combines Python cleaning, MySQL analytical views and linked Tableau dashboards. The analysis tables preserve 99,441 unique order records and 112,650 unique order-item records; category sales reconcile to order-level sales.

Sales. Delivered item value excludes freight. AOV = item sales / distinct delivered orders. Growth compares January-August in both years; concentration uses the full reference period.

Delivery. There are 96,203 valid delivered orders and 6,531 late orders. Eight delivered orders lack an actual delivery date and are excluded from delivery metrics. Late delivery means arrival after the estimated delivery calendar date. Rates use delivery_analysis_valid and is_late. Purchase-month comparisons describe order cohorts.

Reviews. The analysis uses the order-level score represented in the analysis table. Low ratings are 1 or 2 stars. After-delivery eligibility uses review_answer_timestamp on or after actual delivery; missing timestamps are excluded.

Limits

Historical, observational results do not establish causes or current performance. Delivered-order selection, end-of-period maturation, missing data and multiple-review consolidation may affect estimates. Comparisons are not adjusted for seller, category, geography or customer mix. Proposed interventions have not been implemented or evaluated.

Source and reproducibility

Olist Brazilian E-Commerce Public Dataset on Kaggle (https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce). Calculations use the order-summary and order-item CSVs packaged in Olist_Portfolio_Final_v5.twbx. Dashboard filters support exploration; this brief reports a fixed national scope. Values are rounded independently.